

Topic

✓ 1) Operating System [ชุดโปรแกรมที่ควบคุมการทำงานของ Hardware]

- ✓ 1.1) Starting Computer and Mobile Devices
- ✓ 1.2) Shutting Down Computer and Mobile Devices
- ✓ 1.3) Providing a user interface
- ✓ 1.4) Managing Programs
- ✓ 1.5) Managing Memory
 - ✓ - Coordinating tasks
- ✓ 1.6) Configuring devices
- ✓ 1.7) Monitoring performance
- ✓ 1.8) Establishing an Internet Connection
- ✓ 1.9) Providing File, Disk, and System Management Tools
- ✓ 1.10) Updating Operating System software
- ✓ 1.11) Controlling a Network
- ✓ - Type of OS
- ✓ - Desktop OS
 - ✓ - Windows, Mac OS, UNIX, LINUX, Chrome OS
- ✓ - Mobile OS
 - ✓ - Android, iOS, Windows Phone

✓ 2) Programs and APPs

- ✓ 1) Programs and APPs
 - ✓ - OS
- ✓ 2) Obtaining Software
 - ✓ - Retail Software, Custom Software, A web app, A mobile web app, Shareware, Freeware, Open source software, Public-domain Software
- ✓ 3) Categories of Programs and APPs
- ✓ 4) Productivity Applications
 - ✓ - Productivity Applications
 - ✓ - Developing Project
 - ✓ 4.1) Word Processing
 - ✓ 4.2) Presentation
 - ✓ 4.3) Spreadsheet
 - ✓ 4.4) Database
 - ✓ 4.5) Note Taking
 - ✓ 4.6) Calendar and Contact Management
 - ✓ 4.7) Software Suite
 - ✓ 4.8) Project Management
 - ✓ 4.9) Accounting
 - ✓ 4.10) Personal Finance
 - ✓ 4.11) Legal
 - ✓ 4.12) Tax Preparation
 - ✓ 4.13) Document Management
 - ✓ 4.14) Enterprise Computing
- ✓ 5) Graphics and Media Applications
 - ✓ 5.1) Computer-aided design (CAD)
 - ✓ 5.2) Desktop Publishing

- ✓ 5.3) Paint / Image Editing
- ✓ 5.4) Photo Editing and Photo Management
- ✓ 5.5) Video and Audio Editing
- ✓ 5.6) Multimedia and Website Authoring
- ✓ 5.7) Media Player
- ✓ 5.8) Disc Burning
- ✓ 6) Personal Interest Applications
- ✓ 7) Communications Applications
- ✓ 8) Security Tools — Virus
 - ✓ 8.1) Antivirus program
 - ✓ 8.2) Personal firewall
 - ✓ 8.3) Spyware and Adware
 - ✓ 8.4) Internet filter
- ✓ 9) File Disk and System Management Tools
- ✓ 3) Database
 - ✓ 1) Database, Data, and Information
 - ✓ 2) The Hierarchy of Data
 - ✓ 3) File Maintenance
 - ✓ 4) Validating Data
 - ✓ 5) File Processing Systems and Databases
 - ✓ 6) Types of Databases
 - ✓ 7) Database Management System
- ✓ 4) Ethical, Social, and Legal Aspects of IT
 - ✓ 1) Software Theft, Information and Hardware
 - ✓ 2) Information Privacy
 - ✓ 3) Principles of Technology Ethics
 - ✓ 4) Ethical and Social Aspect of IT
 - ✓ 5) Legal Aspects of IT
- ✓ 5) Working in the Enterprise: Systems, Certifications, and Careers
 - ✓ 1) The Technology Industry
 - ✓ 2) Information System in the Enterprise
 - ✓ 3) Technology Careers
 - ✓ 4) Technology Certifications
 - ✓ 5) Job Searching and Career Planning
 - ✓ 6) Creating a Professional Online Presence
- 6) Internet
 - 1) What is Internet?
 - “nuts and bolts” view # Slide Recap Week 12 P.2
 - a service view
 - 2) What's a protocol?
 - 3) Network Edge
 - 4) Access Networks and Physical Media
 - digital subscriber line (DSL)
 - cable network
 - cable network HFC

- FTTH (Fiber to the home) Internet access
 - Enterprise access networks (Ethernet)
 - Wireless access networks
 - data center networks
 - Host: sends packets of data
 - Physical Media / Physical Media: coax, fiber / Physical Media: radio
- 5) The network core
- 6) Internet structure
- a "network of networks"
- 7) Network security
- Bad guys: fake identity / Bad guys: denial of service
 - Line of defense

7) Computer Network

- 1) Network Topology
- A fully connected mesh topology (5 devices)
 - A star topology connection (4 stations)
 - A bus topology connection (3 stations)
 - A ring topology connection (6 stations)
 - A hybrid topology: a star backbone with (3 bus) networks
- 2) Logical Topology (Multiple Access Protocol)
- CSMA
 - collision in CSMA
 - CSMA/CD
 - Token Passing
- 3) Network Communication Standard
- Ethernet
 - Standard Ethernet
 - Change in a standard
 - Fast Ethernet
 - Gigabit Ethernet
 - Ten-Gigabit Ethernet
- 4) Connecting LAN
- Connecting devices
 - Repeater
 - Bridge or Layer 2 switch
 - Router or Layer 3 switch
 - Backbone Network
 - Bus backbone
 - Star backbone
- 5) Logical Addressing
- IP Version 4 Address
 - IPv4 Address Component
 - Network ID / Host ID
 - Getting IPv4 Address
 - IP Address and Subnet Mask
 - Network ID

- การสื่อสารในระบบคอมพิวเตอร์ # P.62-65
- 6) Types of Communication in an IPv4 Network
 - 3 types
 - Unicast / Broadcast / Multicast
 - Special Address
- 7) Class of IPv4
 - Classful IP Addressing - class C
 - Classful IP Addressing - class B
 - Classful IP Addressing - class A
 - Waste of IP Address Space
 - Address Depletion
- การให้บริการบนเครือข่าย

8) Wireless Network

- 1) Element of a wireless network
- 2) Wireless Link Characteristics
 - fading (attenuation) / multipath
 - SNR
 - hidden terminal
 - Code Division Multiple Access (CDMA)
- 3) IEEE 802.11 Wireless LAN
 - 802.11 LAN architecture
 - " channels, association
 - " passive/active scanning
 - " multiple access
 - IEEE 802.11 MAC Protocol: CSMA/CA
 - Avoiding collisions
 - 802.11: advance capabilities
 - 802.15: personal area network (PAN)

9) Information System

- 1) Demand for Digital Skill
- 2) Information System
- 3) System Development
 - Planning Phase
 - Analysis phase
 - Design Phase
 - Implementation
 - support and security phase

✓ 10) Focus on Web Technology

- ✓ 1) Tools for Creating a Website
- ✓ 2) Website Technologies
- ✓ 3) Structure of a Webpage
- ✓ 4) Creating the index.html File
- ✓ 5) HTML with CSS (Cascading Style Sheet)
- ✓ 6) การพัฒนา web client-server
- ✓ 7) การใช้งานเว็บ Web Browser
- ✓ 8) Javascript

✓ 9) Single Page Application

✓ 10) Introduction to API

✓ — Rest API

✓ 11) GraphQL